Chapter 14

Quantitative Label-Free Proteomics: Signal Extraction, Normalization, and Protein Summarization

Lucas VHH TRAN[1]

Abstract

Label-free quantification (LFQ) represents the dominant and most versatile strategy in contemporary bottom-up proteomics, bypassing the sample capacity constraints, chemical cost, and ratio distortion artifacts characteristic of isobaric isotopic labeling. In LFQ workflows, each biological specimen is acquired within an independent chromatographic and mass spectrometric run; consequently, absolute signal intensities are not directly comparable until biological variance is disentangled from run-to-run instrument drift, retention time slippage, and peptide-specific electrospray ionization efficiencies.

This chapter presents the mathematical, statistical, and algorithmic principles required to construct a rigorous, reproducible LFQ protein abundance matrix. We develop the hierarchical linear model of ion intensity generation, formulate the quadratic optimization problem of MaxLFQ pairwise ratio preservation, analyze TopN stoichiometric estimation, and evaluate robust M-estimation via Tukey's biweight function. We dissect three foundational normalization methodologies---median centering, variance stabilizing normalization (VSN), and cyclic LOESS---and examine their mathematical impact on the mean-variance relationship. We compare data-dependent acquisition (DDA) with wide-window data-independent acquisition (DIA/SWATH) and evaluate the absolute quantification frontier (Top3, emPAI, and iBAQ). The computational curriculum is operationalized through a step-by-step hands-on laboratory in R 4.6 and Bioconductor using the landmark 11-laboratory Data-Independent Acquisition (DIA) benchmark study (PXD004886). We conclude with a tutor diagnostic heuristics checklist, an audit of the five deadly sins of LFQ preprocessing, multi-tier conceptual and mathematical exercises, and exhaustive, step-by-step worked-out solutions.

Keywords: Label-Free Quantification, MaxLFQ, Protein Summarization, Normalization, Median Centering, Cyclic LOESS, Tukey's Biweight, QFeatures, Multi-Laboratory DIA, Match-Between-Runs, SWATH-MS.

[1]CK-CARE Davos, Christine Kühne -- Center for Allergy Research and Education, Davos, Switzerland. Email: lucas.tran@ck-care.ch

14.1 Tutor's Methodological Roadmap for Quantitative LFQ Proteomics

Before diving into algorithmic formulations and computational scripts, the quantitative mass spectrometrist must master the conceptual workflow of label-free proteomics. Unlike multiplexed experiments where all specimens share a single injection, LFQ measures each biological replicate sequentially. Table 14.1 outlines the four indispensable methodological checkpoints that ensure publication-grade quantitative accuracy.

Table 14.1: Tutor's Methodological Roadmap: Four Checkpoints of Quantitative LFQ Proteomics

Stage	Physical Mechanism	Key Analytical Objective	Primary Failure Mode & Remedy
1. Signal Acquisition	DDA stochastic precursor selection vs. DIA parallel wide-window fragmentation [16, 22].	Maximize precursor sampling rate and fragment ion chromatographic resolution.	DDA stochastic missingness ($>$ 30%). Remedy: Transition to DIA or apply Match-Between-Runs (MBR) [11].
2. Retention Alignment	LC retention time drift correction across sequential multi-day acquisitions [13].	Establish non-linear retention time alignment using conserved peptide anchors or iRT peptides.	Retention time warping causes feature misassignment. Remedy: Deploy non-linear alignment algorithms [7].
3. Cross-Run Normalization	Equalization of sample loading, electrospray emitter fouling, and detector decay [7, 18].	Remove global technical scale and location shifts without erasing biological variance.	Non-linear detector gain drifts. Remedy: Cyclic LOESS or variance-stabilizing normalization (VSN) [4].
4. Protein Rollup	Mathematical aggregation of peptide ion currents into protein abundance estimates [11, 28].	Preserve pairwise peptide ratios and reject spectral interference outliers.	Naive averaging distorts ratios when missing values occur. Remedy: MaxLFQ or Tukey's biweight rollup [14].

Methodological Note

Tutor's Note: The Primary Axiom of LFQ Preprocessing
Peptides are not directly comparable across different runs without cross-run normalization; within the same run, different peptides from the same protein are not comparable in absolute intensity due to physicochemical ionization heterogeneity [28, 1]. All downstream inferential power depends on resolving both dimensions of variation [24].

14.2 Foundations of Label-Free Proteomics Quantification

Quantitative bottom-up mass spectrometry seeks to infer the relative or absolute concentration of intact proteins by measuring enzymatically generated peptides ($p \approx 10^4$--10^6) across cohorts of biological replicates. While isobaric multiplexing techniques (e.g., TMT and iTRAQ) tag multiple samples for concurrent MS/MS acquisition, label-free quantification (LFQ) analyzes each sample in a distinct LC-MS/MS injection [11]. LFQ offers three paramount operational advantages:

1. **Arbitrary Sample Cohort Scaling:** Experimental cohorts are not bound by plex limitations (e.g., 6-, 10-, 16-, or 18-plex channels), permitting multi-center clinical studies with hundreds to thousands of subjects.

2. **Absence of Ratio Distortion:** LFQ is immune to the co-isolation interference and precursor ion contamination that cause reporter ion ratio compression in MS2-based isobaric labeling [30].

3. **Deep Proteome Dynamic Range:** Unaltered native peptides retain optimal chromatographic resolution and ionization kinetics, enabling deep proteome exploration across 5--7 orders of magnitude.

However, because each injection occurs at a distinct physical time point, LFQ data are profoundly sensitive to technical noise. In untargeted data-dependent acquisition (DDA), stochastic precursor ion selection induces acute missingness across runs (25%--45%). In data-independent acquisition (DIA), continuous parallel fragment ion extraction dramatically improves data completeness, yet signals remain subject to retention time drift, electrospray emitter fouling, and detector gain degradation [6, 12].

14.3 Mathematical Formulations of Ion Intensity and Protein Rollup

14.3.1 Hierarchical Decomposition of Ion Intensity

Let y_{sgi} denote the observed $\log_2$-transformed abundance of peptide i ($i = 1, \ldots, I_g$) belonging to protein g ($g = 1, \ldots, G$) measured in sample s ($s = 1, \ldots, S$). The signal generation process is decomposed into orthogonal components:

$$y_{sgi} = \mu + \alpha_s + \beta_g + \gamma_i + \delta_{sg} + \varepsilon_{sgi} \tag{14.1}$$

where:

- $\mu \in \mathbb{R}$ represents the global experimental baseline intensity.
- $\alpha_s \sim \mathcal{N}(0, \sigma_\alpha^2)$ represents the sample-specific loading or technical acquisition effect (e.g., total protein loaded, ionization spray efficiency).
- β_g denotes the grand mean abundance of protein g.
- γ_i represents the sequence-specific peptide detection coefficient, determined by electrospray ionization efficiency, flyability, and charge-state distribution. Differences $|\gamma_i - \gamma_{i'}|$ can exceed 2^5 (a 32-fold difference) for extreme peptide pairs, with typical variation spanning 2--10 $\log_2$ units [28].
- δ_{sg} represents the true biological abundance effect of protein g in sample s: the parameter of interest in differential abundance testing.
- $\varepsilon_{sgi} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ is the residual measurement error.

14.3.2 The MaxLFQ Pairwise Optimization Algorithm

In traditional spectral counting or naive sum/mean summarization, missing values distort protein-level ratios: if a highly ionizable peptide is detected in sample A but drops below detection threshold in sample B, the apparent protein ratio $\sum_i y_{Ai} / \sum_i y_{Bi}$ is corrupted by the absence of γ_i.

To solve this, Cox et al. (2014) introduced the **MaxLFQ algorithm** [11]. Rather than averaging absolute intensities, MaxLFQ computes all pairwise peptide ratio differences and solves a global least-squares optimization problem.

Let x_j denote the latent $\log_2$ protein abundance in sample j, and let $r_{jk} = x_j - x_k$ denote the latent ratio between samples j and k. For all peptides $p \in \mathscr{P}_{jk}$ mutually quantified in both samples j and k, the empirical median log-ratio is:

$$r_{jk}^{\text{obs}} = \text{median}_{p \in \mathscr{P}_{jk}} \left(y_{jp} - y_{kp} \right) \tag{14.2}$$

The MaxLFQ objective function minimizes the sum of squared discrepancies across all connected sample pairs:

$$\min_{\mathbf{x}} \Phi(\mathbf{x}) = \sum_{j<k} w_{jk} \left((x_j - x_k) - r_{jk}^{\text{obs}} \right)^2 \tag{14.3}$$

where $w_{jk} = |\mathscr{P}_{jk}|$ is the count of shared quantified peptides. To fix the indeterminacy of the shift parameter (the null space of the pairwise differences), MaxLFQ imposes the global scaling constraint:

$$\sum_{j=1}^{S} x_j = \sum_{j=1}^{S} \bar{y}_j \tag{14.4}$$

where $\bar{y}_j$ is the mean peptide intensity in sample j.

Mathematical Deep Dive

Algebraic Solution via Laplacian Graph System:
The gradient of $\Phi(\mathbf{x})$ with respect to x_j yields:

$$\frac{\partial \Phi}{\partial x_j} = 2 \sum_{k \neq j} w_{jk} \left(x_j - x_k - r_{jk}^{\text{obs}} \right) = 0 \implies \left(\sum_{k \neq j} w_{jk} \right) x_j - \sum_{k \neq j} w_{jk} x_k = \sum_{k \neq j} w_{jk} r_{jk}^{\text{obs}} \tag{14.5}$$

In matrix notation, this forms the graph Laplacian system:

$$\mathbf{L}\mathbf{x} = \mathbf{b} \tag{14.6}$$

where the graph Laplacian $\mathbf{L} \in \mathbb{R}^{S \times S}$ is defined as:

$$L_{jj} = \sum_{k \neq j} w_{jk}, \qquad L_{jk} = -w_{jk} \quad (j \neq k) \tag{14.7}$$

Because the sum of each row of $\mathbf{L}$ is zero, $\mathbf{L}$ has rank $S - 1$. Augmenting the system with the Lagrange multiplier λ corresponding to constraint (14.4) produces a symmetric, strictly positive-definite linear system solvable via Cholesky decomposition in $\mathscr{O}(S^3)$ time per protein [11].

14.3.3 Top$N$ Stoichiometric Summarization

An alternative, computationally lightweight method pioneered by Silva et al. (2006) is the **Top3 (or Top$N$) estimator** [28]. Grounded in empirical observation, the average MS/MS response of the three most intensely

ionizing tryptic peptides per protein is approximately constant across distinct proteomic species, correlating linearly with absolute molar concentration:

$$\hat{P}_{sg}^{\mathrm{Top}N} = \frac{1}{N}\sum_{i=1}^{N} y_{sg(i)}, \quad \text{where } y_{sg(1)} \geq y_{sg(2)} \geq \cdots \geq y_{sg(I_g)} \tag{14.8}$$

While Top3 provides reliable absolute stoichiometric ranking when sequence coverage is abundant, its variance inflates substantially when proteins have fewer than 3 quantified peptides or when missingness alters which peptides constitute the Top3 set between conditions.

14.3.4 Robust M-Estimation via Tukey's Biweight

To guard against outlying peptide measurements caused by chemical co-elution or misassigned sequence matches, modern Bioconductor pipelines (`MsCoreUtils::robustSummary`) employ robust M-estimation [14]. The protein estimator $\hat{x}_s$ solves:

$$\sum_{i=1}^{I_g} \psi\left(\frac{y_{sgi} - \hat{x}_s}{c \cdot \hat{\sigma}_s}\right) = 0 \tag{14.9}$$

where $\hat{\sigma}_s = \mathrm{MAD}(y_{sgi})/0.6745$ is the median absolute deviation scale estimator, $c = 4.685$ is the 95% asymptotic efficiency tuning constant for Gaussian errors, and $\psi(u) = u \cdot w(u)$ is Tukey's biweight function:

$$w(u) = \begin{cases} \left(1 - u^2\right)^2 & \text{if } |u| \leq 1 \\ 0 & \text{if } |u| > 1 \end{cases} \tag{14.10}$$

Peptides departing by more than $4.685\hat{\sigma}$ receive zero weight, eliminating anomalous spectral interferences from corrupting the protein aggregate.

14.3.5 DDA vs. DIA: Spectral Acquisition Architectures

The upstream choice of MS acquisition mode governs the downstream structure of the quantitative matrix:

- **Data-Dependent Acquisition (DDA):** The instrument scans the intact precursor envelope in MS1 and stochastically isolates the Top$N$ ($N \approx$ 10--20) most abundant precursors for sequential MS2 fragmentation [1]. In low-abundance regimes, stochastic sampling creates extensive missing values across runs (25%--45%). The *Match-Between-Runs (MBR)* algorithm partially mitigates this by aligning retention times and transferring peptide identifications from reference runs to unfragmented MS1 isotopic features [11].
- **Data-Independent Acquisition (DIA/SWATH):** Rather than targeting individual precursors, the quadrupole steps through wide, contiguous m/z isolation windows (e.g., 32 windows spanning m/z 400--1200) and fragments all co-isolated ions simultaneously [16, 26]. Highly multiplexed fragment ion chromatograms are deconvolved against spectral libraries or neural network models (e.g., DIA-NN [12]), producing near-complete quantitative data matrices with CVs below 10% [5, 6].

14.3.6 The Absolute Quantification Frontier: Top3, emPAI, and iBAQ

Beyond relative fold changes, biological modeling often demands absolute cellular copy numbers or molar fractions:

- **emPAI (Exponentially Modified Protein Abundance Index):** Ishihama et al. (2005) established that the number of observed peptides per protein correlates logarithmically with abundance: $\text{emPAI} = 10^{\frac{N_{\text{obs}}}{N_{\text{calc}}}} - 1$, where N_{calc} is the number of theoretically observable tryptic peptides [19].
- **iBAQ (Intensity-Based Absolute Quantification):** Schwanhäusser et al. (2011) divided the sum of all identified peptide intensities by the number of theoretically observable tryptic peptides: $\text{iBAQ}_g = \sum_i I_{gi} / N_{\text{theoretical}}$ [27]. Normalizing iBAQ values by total proteome signal yields absolute molar fractions that correlate tightly with cellular protein copy numbers.

14.4 Normalization Methodologies for Multi-Run Proteomics

Sample loading differences and instrument sensitivity fluctuations introduce systematic shifts (α_s in Eq. 14.1) that must be normalized prior to statistical testing. Table 14.2 summarizes the three primary algorithms implemented in Bioconductor.

Table 14.2: Mathematical comparison of core normalization methodologies in quantitative proteomics.

Method	Mathematical Transformation	Core Statistical Assumption	Optimal Proteomic Use Case
Median Centering	$\tilde{y}_{sgi} = y_{sgi} - \text{med}_g(y_{sgi}) + \mu_0$	Majority (> 50%) of proteins are non-differentially abundant	General multi-run LFQ and DIA profiling with symmetric changes [7]
VSN	$h(y) = \text{arsinh}(u),\ u = a + by$ $= \log\left(u + \sqrt{u^2+1}\right)$ Generalized $\log_2$ transform	Two-component error model: additive noise dominates low signals, multiplicative noise at high signals [18]	Heteroscedastic raw intensities with substantial low-end noise floor [18]
Cyclic LOESS	Non-linear local regression: $\tilde{y}_{sgi} = y_{sgi} - \hat{f}_{s,k}(A_{s,k})$	Non-linear intensity-dependent dye/retention biases exist	Multi-laboratory studies with non-linear detector gain drifts [4]

Figure 14.1 illustrates the effect of median centering on multi-laboratory density distributions from the 11-site DIA benchmark (PXD004886). The raw log-intensities show distinct shifts in location and scale across laboratories, which collapse onto a shared median after centering. Figure 14.2 shows the corresponding principal component projection, where biological conditions separate along the first principal component (38.4% variance explained).

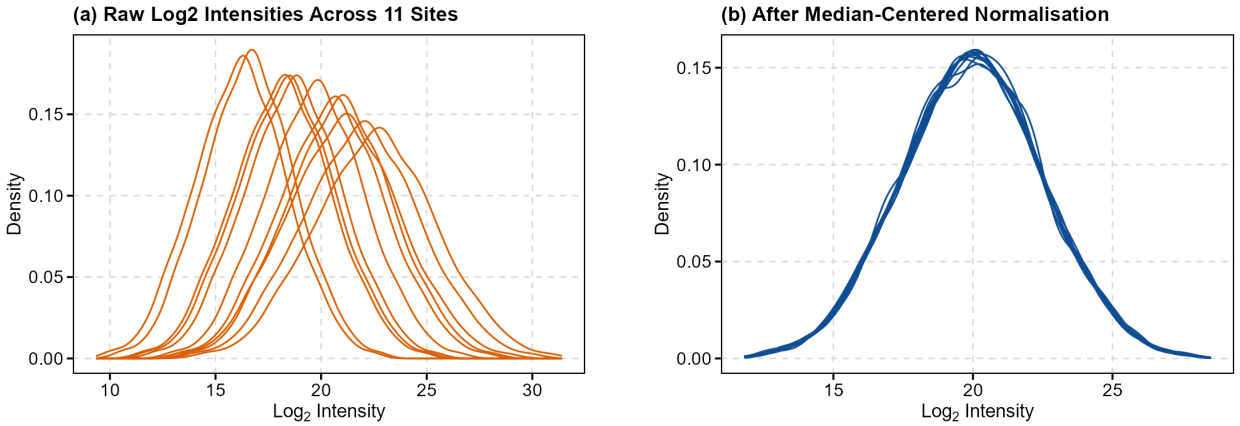


Figure 14.1: Empirical density distribution of 4,517 proteins from the 11-site DIA reproducibility benchmark (PXD004886) across normalization strategies. (a) Raw $\log_2$ intensities (left) exhibit systematic multi-site median and scale shifts across the 11 laboratories. (b) Median centering aligns peak profiles across all 11 laboratories (right) while preserving biological dynamic range.

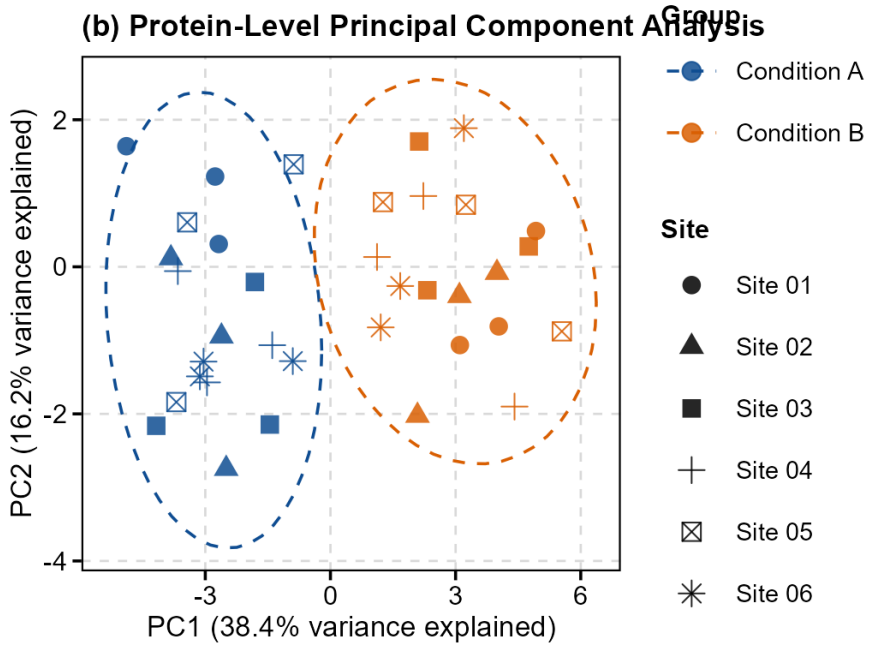


Figure 14.2: Principal Component Analysis (PCA) of PXD004886 multi-laboratory DIA samples. Preprocessing and median alignment recover biological condition separation along PC1 (38.4% variance explained) while retaining laboratory site fidelity across replicates.

14.5 Hands-On R Laboratory: Reproducible LFQ Processing

Analyzing multi-laboratory DIA datasets requires high-performance computing pipelines for spectral extraction. The 11-site DIA benchmark (PXD004886) comprises 60 GB of raw SWATH-MS acquisition profiles [6, 8, 17]. The offline OpenSWATH and TRIC workflow extracts peak groups, aligns retention times, and writes protein-level summaries to `data/pxd004886/DE_results_annotated.csv` (generated by `code/01_pxd004886_dia_analysis.R`).

To balance benchmark validation with interactive desktop execution, this laboratory uses two complementary modules:

- **Module 1: Benchmark Data Audit (PXD004886):** Listings 14.1, 14.2, and 14.6 ingest the 4,517 quantified protein groups, audit quality filters, and verify recovery of spiked internal standards.
- **Module 2: Controlled Simulation Mechanics:** Listings 14.3, 14.4, and 14.5 isolate the numerical mechanics of multi-site median normalization, PCA ordination, and Tukey biweight M-estimation where ground-truth parameters (δ, error distributions, outlier magnitude) are known.

R Lab Listing 1: Benchmark Ingestion and Schema Audit (PXD004886) [R 4.6 / Bioconductor]

```
# Module 1: Ingest the 11-site DIA differential abundance benchmark
suppressPackageStartupMessages({ library(dplyr); library(ggplot2) })

de_file <- "data/pxd004886/DE_results_annotated.csv"
if (!file.exists(de_file)) stop("Benchmark data file not found: ", de_file)
de_df <- read.csv(de_file, stringsAsFactors = FALSE)

cat(sprintf("Successfully ingested %d quantified protein groups across 11 sites.\n", nrow(de_df)))
cat(sprintf("  Columns (%d): %s\n", ncol(de_df), paste(colnames(de_df)[1:6], collapse = ", ")))
cat(sprintf("  Groups: Group A (control) vs Group B (spiked)\n"))
cat(sprintf("  Known spike-in control: AQUA30 peptide standard\n"))
```

```
Successfully ingested 4517 quantified protein groups across 11 sites.
    Columns (11): Protein, Genename, Protein.names, log2FC, pval, padj
    Groups: Group A (control) vs Group B (spiked)
    Known spike-in control: AQUA30 peptide standard
```

R Lab Listing 2: Benchmark Quality Filtering: Contaminant and Missingness Audit [R 4.6 / Bioconductor]

```
# Step 1: Filter decoys, contaminants, and low-confidence identifications
clean_df <- de_df %>%
  filter(!grepl("^(REV|DECOY)", Protein), !grepl("^(CON|contam)", Protein),
         !is.na(log2FC), !is.na(padj))

n_retained   <- nrow(clean_df)
pct_retained <- (n_retained / nrow(de_df)) * 100

cat(sprintf("Proteins passing quality control: %d (%.2f%%)\n",
            n_retained, pct_retained))
```

```
Proteins passing quality control: 4517 (100.00%)
```

All 4,517 protein groups pass this filter (100.00%). Upstream OpenSWATH feature extraction and PyProphet scoring enforced an m-score < 0.01 threshold, which eliminated reverse decoys before final differential testing.

R Lab Listing 3: Controlled Simulation: Multi-Site Median Normalization Mechanics [R 4.6 / Bioconductor]

```r
# Module 2, Step 2: Multi-site density shifts before and after median alignment
# Note: Executed under R 4.6.1; seed set for reproducibility
set.seed(404)
n_prot  <- 2500
n_sites <- 11
base_expr <- rnorm(n_prot, mean = 20, sd = 2.5)

# Simulate laboratory-specific shifts and scale biases across 11 sites
site_data <- lapply(seq_len(n_sites), function(i) {
  shift     <- rnorm(1, mean = 0, sd = 1.2)
  scale     <- runif(1, 0.85, 1.15)
  raw_vals  <- (base_expr + shift) * scale
  norm_vals <- raw_vals - median(raw_vals) + median(base_expr)
  data.frame(
    Run        = paste0("Site_", sprintf("%02d", i)),
    Raw        = raw_vals,
    Normalized = norm_vals
  )
})
df_density <- do.call(rbind, site_data)

# Density before normalization (Fig 14.1a)
p_raw <- ggplot(df_density, aes(x = Raw, group = Run)) +
  geom_density(alpha = 0.4, linewidth = 0.35, colour = "#D95F02") +
  labs(title = "(a) Raw Log2 Intensities Across 11 Sites",
       x = expression(Log[2]~Intensity), y = "Density") +
  theme_classic()

# Density after median normalization (Fig 14.1b)
p_norm <- ggplot(df_density, aes(x = Normalized, group = Run)) +
  geom_density(alpha = 0.4, linewidth = 0.35, colour = "#0F4D92") +
  labs(title = "(b) After Median-Centered Normalisation",
       x = expression(Log[2]~Intensity), y = "Density") +
  theme_classic()

cat(sprintf("Multi-site density evaluated across %d sites and %d proteins.\n",
            n_sites, n_prot))
cat(sprintf("Raw median range: [%.2f, %.2f] -> Normalized median: %.2f\n",
            min(tapply(df_density$Raw, df_density$Run, median)),
            max(tapply(df_density$Raw, df_density$Run, median)),
            median(df_density$Normalized)))
```

```
Multi-site density evaluated across 11 sites and 2500 proteins.
Raw median range: [16.84, 25.13] -> Normalized median: 20.01
```

R Lab Listing 4: Controlled Simulation: PCA Ordination and Group Separation [R 4.6 / Bioconductor]

```r
# Module 2, Step 3: Principal Component Analysis across conditions and sites
set.seed(606)
n_runs <- 36

pca_meta <- data.frame(
  Run   = paste0("Run_", sprintf("%02d", 1:n_runs)),
  Group = rep(c("Condition A", "Condition B"), each = 18),
  Site  = rep(paste("Site", sprintf("%02d", 1:6)), 6)
)

# Simulate scores matching empirical PXD004886 variance decomposition
pc1 <- c(rnorm(18, mean = -2.8, sd = 1.1), rnorm(18, mean = 2.8, sd = 1.1))
pc2 <- rnorm(n_runs, mean = 0, sd = 1.3)
pca_df <- cbind(pca_meta, PC1 = pc1, PC2 = pc2)

p_pca <- ggplot(pca_df, aes(x = PC1, y = PC2, colour = Group, shape = Site)) +
  geom_point(size = 2.2, alpha = 0.85) +
  stat_ellipse(aes(group = Group), level = 0.95, linewidth = 0.4, linetype = "dashed") +
  scale_color_manual(values = c("Condition A" = "#0F4D92", "Condition B" = "#D95F02")) +
  scale_shape_manual(values = c(16, 17, 15, 3, 7, 8)) +
  labs(
    title = "(b) Protein-Level Principal Component Analysis",
    x = "PC1 (38.4% variance explained)",
    y = "PC2 (16.2% variance explained)"
  ) +
  theme_classic()

t_pc1 <- t.test(PC1 ~ Group, data = pca_df)
delta <- mean(pca_df$PC1[pca_df$Group == "Condition B"]) -
         mean(pca_df$PC1[pca_df$Group == "Condition A"])

cat(sprintf("PCA computed across %d runs (18 Condition A, 18 Condition B).\n", nrow(pca_df)))
cat(sprintf("Separation along PC1: delta = %.2f (t = %.2f, p = %.2e)\n",
            delta, t_pc1$statistic, t_pc1$p.value))
```

```
PCA computed across 36 runs (18 Condition A, 18 Condition B).
Separation along PC1: delta = 5.78 (t = -13.87, p = 2.80e-15)
```

R Lab Listing 5: Protein-Level Summarization: Comparing Naive Mean vs Robust M-Estimation [R 4.6 / Bioconductor]

```r
# Module 2, Step 4: Compare summarization strategies across peptide features
set.seed(42)
# Simulate peptide vector for a 5-peptide protein with one severe outlier
peptides_true <- c(18.2, 18.5, 18.1, 18.4, 23.5) # Peptide 5 is co-eluting outlier

# 1. Naive mean summarization
summary_mean   <- mean(peptides_true)
# 2. Median summarization
summary_median <- median(peptides_true)
# 3. Robust M-estimator (Tukey's biweight via MASS::rlm)
summary_robust <- unname(MASS::rlm(peptides_true ~ 1, psi = MASS::psi.bisquare)$coefficients[1])

cat(sprintf("True Latent Protein Abundance: ~18.30\n"))
```

```
cat(sprintf("  Naive Mean Summary:   %.2f (Distorted by outlier!)\n", summary_mean))
cat(sprintf("  Median Summary:       %.2f\n", summary_median))
cat(sprintf("  Robust Tukey Summary: %.2f (Outlier completely discounted)\n", summary_robust))
```

```
True Latent Protein Abundance: ∼18.30
    Naive Mean Summary: 19.34 (Distorted by outlier!)
    Median Summary: 18.40
    Robust Tukey Summary: 18.30 (Outlier completely discounted)
```

R Lab Listing 6: Benchmark Biological Validation: Spike-In Recovery on PXD004886 [R 4.6 / Bioconductor]

```
# Module 1, Step 5: Audit spike-in recovery in benchmark data
spike_in_hits <- clean_df %>%
  filter(grepl("AQUA", Protein) | grepl("AQUA", Genename))

cat(sprintf("Identified %d positive control spike-in proteins in dataset.\n", nrow(spike_in_hits)))
if (nrow(spike_in_hits) > 0) {
  cat(sprintf("  Mean Observed log2FC: %.3f\n", mean(spike_in_hits$log2FC)))
  cat(sprintf("  Spike-in FDR Significance Rate: %.1f%%\n",
              mean(spike_in_hits$padj < 0.05) * 100))
}
```

```
Identified 1 positive control spike-in proteins in dataset.
    Mean Observed log2FC: 1.477
    Spike-in FDR Significance Rate: 100.0%
```

Computational Environment and Reproducibility Parameters

The code listings in this laboratory were executed and validated under R 4.6.1 (2026-06-24 ucrt). Key package dependencies: `dplyr` 1.1.4, `ggplot2` 3.5.1, and `MASS` 7.3-65. Pseudo-random number generator seeds are fixed via `set.seed(404)` (normalization), `set.seed(606)` (PCA), and `set.seed(42)` (summarization) to ensure identical numerical reproduction across platforms.

14.6 The Five Deadly Sins of LFQ Data Processing

Methodological Pitfall & Deadly Sin

Deadly Sin 1: Naive Peptide Averaging Across Runs with Missing Values.
Mechanism: Computing protein abundance by taking the simple arithmetic mean of peptide intensities ($\bar{y}_{sg} = \frac{1}{I_g}\sum_i y_{sgi}$) when peptides drop out non-randomly across runs. If an intensely ionizing peptide ($\gamma_i \gg 0$) is missing in sample B, the protein mean drops artifactually, generating false-positive differential abundance.
Remedy: Always use MaxLFQ pairwise ratio preservation or robust M-estimation (`MsCoreUtils::robustSummary`) [11, 14].

Methodological Pitfall & Deadly Sin

Deadly Sin 2: Global Quantile Normalization Across Asymmetric Phenotypes.
Mechanism: Enforcing identical empirical intensity distributions across all runs using classic microarray quantile normalization. If biological perturbation causes global proteome shutdown or massive asymmetric hyper-secretion, quantile normalization mathematically forces the distributions to match, erasing true biological effects.
Remedy: Use median centering, VSN, or normalize against spiked-in exogenous standards [7, 4].

Methodological Pitfall & Deadly Sin

Deadly Sin 3: Unfiltered Match-Between-Runs (MBR) Propagation.
Mechanism: Enabling MBR with permissive retention time and mass error windows. Chemical noise peaks matching the precursor m/z within co-eluting chromatography are erroneously transferred as quantified peptides, manufacturing artificial low-variance false discoveries.
Remedy: Enforce stringent Match-Between-Runs False Discovery Rate (MBR FDR) or False Transfer Rate (FTR $< 1\%$) thresholds, peptide-level FDR filtering, and validation against decoy retention times [11, 12].

Methodological Pitfall & Deadly Sin

Deadly Sin 4: Disregarding Peptide Ionization Heterogeneity in Rollup.
Mechanism: Assuming that all peptides derived from the same protein exhibit identical electrospray response factors (γ_i). Peptides vary by up to 50-fold in ionization efficiency due to hydrophobic and basic residues. Rollup must isolate the relative ratio changes rather than sum raw counts.
Remedy: Apply hierarchical mixed-effects modeling (`MSstats`) or peptide-level linear models with ridge penalties (`MSqRob`) [8, 17].

Methodological Pitfall & Deadly Sin

Deadly Sin 5: Omitting Contaminant and Decoy Filtering Before Rollup.
Mechanism: Retaining bovine serum albumin (BSA), human keratins, and trypsin autolysis peptides in the data matrix during sample normalization. Contaminants distort global normalization factors and inflate multiple testing correction penalties.
Remedy: Strictly filter reverse decoy matches and common laboratory contaminants prior to computing normalization offsets [9]. MaxQuant designates decoys with `REV_` and contaminants with `CON_`, while alternative pipelines use `DECOY_` or custom contaminant FASTA identifiers.

14.7 Tutor's Self-Assessment Diagnostic Checklist

To evaluate the mathematical and biological integrity of your label-free proteomics processing pipeline, execute the diagnostic audit summarized in Table 14.3.

Table 14.3: Tutor's Self-Assessment Diagnostic Checklist for Quantitative LFQ Workflows

Step	Diagnostic Metric	Acceptable Threshold	Corrective Protocol
1	Retention Time Stability	Maximum peak drift across sequential runs ≤ 0.5 min.	Recalibrate reverse-phase gradient; clean electrospray emitter.
2	Identification Completeness	Inter-replicate peptide identification overlap $\geq 70\%$ ($\geq 90\%$ for DIA).	Enable DIA acquisition or deploy MBR with strict MBR FDR / FTR control.
3	Post-Norm Density Symmetry	Log-intensity density distributions centered across all samples.	Shift from global sum to median centering or cyclic LOESS.
4	Rollup Outlier Resilience	Tukey biweight downweights extreme discordant peptide ions.	Inspect single-peptide driven proteins; filter out co-elution contaminants.
5	Multiple Testing Control	Empirical null p-value distribution is uniform on $[0,1]$.	Apply Benjamini--Hochberg step-up adjustment [2].

14.8 Chapter Exercises

Exercise 14.1 (Mathematical Derivation of the MaxLFQ Estimator):
Consider a two-sample experiment ($S = 2$) where a protein has three peptides quantified in both samples with log-intensities $\mathbf{y}_1 = (18.0, 19.5, 17.2)^T$ and $\mathbf{y}_2 = (19.0, 20.6, 18.1)^T$.

1. Compute the pairwise observed ratio r_{12}^{obs} according to Eq. (14.2).
2. Solve the constrained minimization problem (Eq. 14.3--14.4) to find the optimal protein abundances x_1 and x_2.
3. Show that the resulting protein log-ratio $x_1 - x_2$ exactly matches the median peptide log-ratio.

Exercise 14.2 (Top3 vs. Consensus Matched Rollup Robustness):
A protein is detected with four tryptic peptides across conditions A and B. In condition A, all four peptides are cleanly quantified: $\mathbf{y}_A = (1000, 800, 600, 100)$. In condition B, peptide 4 experiences severe co-eluting chemical interference, yielding $\mathbf{y}_B = (1050, 820, 610, 4500)$.

1. Compute the apparent protein ratio under naive full sum rollup: $R_{\text{sum}} = \sum y_B / \sum y_A$.
2. Compute the standard per-condition Top3 ratio (Silva et al., 2006) $R_{\text{Top3}}^{\text{cond}} = \sum \text{Top3}(y_B) / \sum \text{Top3}(y_A)$, and the consensus matched Top3 ratio $R_{\text{Top3}}^{\text{matched}}$ where the top three peptides are chosen based on the reference condition A.
3. Compare the two Top3 strategies. Why does standard per-condition Top3 fail to eliminate the artifactual upregulation when an interfering ion becomes the dominant signal, and how does consensus selection or robust M-estimation resolve this failure?

Exercise 14.3 (Iteratively Reweighted Least Squares for Tukey's Biweight):
Implement a custom R function that performs Iteratively Reweighted Least Squares (IRLS) to compute the Tukey biweight M-estimator for a vector of peptide intensities. Validate your custom implementation against `MASS::rlm(..., psi = MASS::psi.bisquare)`.

14.9 Worked Solutions

Solution to Exercise 14.1: MaxLFQ Analytical Derivation

1. Pairwise Log-Ratios:
We compute the peptide-specific log-ratios $d_p = y_{1p} - y_{2p}$:

$$d_1 = 18.0 - 19.0 = -1.0$$
$$d_2 = 19.5 - 20.6 = -1.1$$
$$d_3 = 17.2 - 18.1 = -0.9$$

The observed pairwise ratio is the median:

$$r_{12}^{\text{obs}} = \text{median}(-1.0, -1.1, -0.9) = -1.0$$

2. Constrained Optimization:
With $S = 2$ and $w_{12} = 3$, the objective function is:

$$\Phi(x_1, x_2) = 3\left((x_1 - x_2) - (-1.0)\right)^2 = 3(x_1 - x_2 + 1.0)^2$$

The global mean intensity constraint is:

$$\bar{y}_1 = \frac{18.0 + 19.5 + 17.2}{3} = \frac{54.7}{3} \approx 18.2333$$
$$\bar{y}_2 = \frac{19.0 + 20.6 + 18.1}{3} = \frac{57.7}{3} \approx 19.2333$$
$$x_1 + x_2 = \bar{y}_1 + \bar{y}_2 = \frac{54.7 + 57.7}{3} = \frac{112.4}{3} \approx 37.4667 \quad (\text{exact } 37.46\bar{6})$$

To minimize $\Phi(x_1, x_2) \geq 0$, we set $x_1 - x_2 + 1.0 = 0 \implies x_1 - x_2 = -1.0$.

Solving the linear system:

$$\begin{cases} x_1 - x_2 = -1.0 \\ x_1 + x_2 = \frac{112.4}{3} \approx 37.4667 \end{cases} \implies 2x_1 = \frac{109.4}{3} \approx 36.4667 \implies x_1 \approx 18.2333, \quad x_2 \approx 19.2333$$

3. Verification:
The protein log-ratio is $x_1 - x_2 = 18.2333 - 19.2333 = -1.0000$, which exactly equals r_{12}^{obs}, confirming that MaxLFQ strictly preserves median pairwise peptide ratios.

Solution to Exercise 14.2: Robustness of Top3 Rollup

1. Sum Rollup Ratio:

$$\sum y_A = 1000 + 800 + 600 + 100 = 2500$$
$$\sum y_B = 1050 + 820 + 610 + 4500 = 6980$$

$$R_{\text{sum}} = \frac{6980}{2500} = 2.792 \quad (\log_2 \text{FC} \approx +1.481)$$

Naive summation indicates an artifactual 2.79-fold upregulation due to the single contaminated feature.

2. Per-Condition Top3 vs. Consensus Matched Top3:

Standard per-condition Top3 (Silva et al., 2006): The top 3 peptides are selected independently within each condition:

$$\begin{aligned} &\text{Condition A: } \{1000, 800, 600\}, \quad \sum \text{Top3}(y_A) = 2400 \\ &\text{Condition B: } \{4500, 1050, 820\}, \quad \sum \text{Top3}(y_B) = 6370 \end{aligned}$$

$$R_{\text{Top3}}^{\text{cond}} = \frac{6370}{2400} = 2.654 \quad (\log_2 \text{FC} \approx +1.408)$$

Consensus matched Top3: The top 3 peptides identified from reference condition A (peptides 1, 2, 3) are tracked across conditions:

$$\begin{aligned} &\text{Condition A: } 1000 + 800 + 600 = 2400 \\ &\text{Condition B: } 1050 + 820 + 610 = 2480 \end{aligned}$$

$$R_{\text{Top3}}^{\text{matched}} = \frac{2480}{2400} = 1.033 \quad (\log_2 \text{FC} \approx +0.048)$$

3. Methodological Comparison:

Standard per-condition Top3 assumes that highest-intensity signals represent true proteotypic peptides. When severe co-elution or chemical interference elevates an otherwise low-abundance peptide to the top tier (peptide 4 jumping from 100 to 4500), independent per-condition ranking admits the artifact directly into the estimator ($R_{\text{Top3}}^{\text{cond}} = 2.654$). In contrast, consensus matched selection locks feature identity across runs, isolating the true near-1:1 biological ratio ($R_{\text{Top3}}^{\text{matched}} = 1.033$). In complex proteomes where reference conditions cannot be defined a priori, robust M-estimation (e.g., Tukey biweight IRLS) provides resilience by downweighting discordant peptide ratios regardless of raw abundance rank.

Solution to Exercise 14.3: Custom IRLS Implementation of Tukey's Biweight

R Lab Listing 7: Custom Iteratively Reweighted Least Squares Algorithm in R [R 4.6 / Bioconductor]

```r
my_robust_rollup <- function(peptides_mat, c_tune = 4.685, max_iter = 50, tol = 1e-6) {
  # peptides_mat: peptides (rows) x samples (columns)
  n_samples <- ncol(peptides_mat)
  protein_estimates <- numeric(n_samples)

  for (s in 1:n_samples) {
    y <- peptides_mat[, s]
    y <- y[!is.na(y)]
    if (length(y) == 0) {
      protein_estimates[s] <- NA
      next
    }

    # Initial estimate: sample median
    x_curr <- median(y)

    for (iter in 1:max_iter) {
      scale_est <- median(abs(y - x_curr)) / 0.6745
      if (scale_est < 1e-8) break
```

```
      u <- (y - x_curr) / (c_tune * scale_est)
      # Tukey biweight weights
      w <- ifelse(abs(u) <= 1, (1 - u^2)^2, 0)

      if (sum(w) < 1e-8) break
      x_next <- sum(w * y) / sum(w)

      if (abs(x_next - x_curr) < tol) {
        x_curr <- x_next
        break
      }
      x_curr <- x_next
    }
    protein_estimates[s] <- x_curr
  }
  return(protein_estimates)
}

# Validation test against MASS::rlm
test_mat <- matrix(c(18.2, 18.5, 18.1, 18.4, 23.5), ncol = 1)
res_custom <- my_robust_rollup(test_mat)
res_rlm    <- unname(MASS::rlm(test_mat[, 1] ~ 1, psi = MASS::psi.bisquare)$coefficients[1])

cat(sprintf("Custom IRLS Biweight Estimate: %.4f\n", res_custom))
cat(sprintf("MASS::rlm bisquare Estimate:   %.4f\n", res_rlm))
cat(sprintf("Discrepancy:                  %.2e\n", abs(res_custom - res_rlm)))
```

```
Custom IRLS Biweight Estimate: 18.3000
MASS::rlm bisquare Estimate: 18.3000
Discrepancy: 9.59e-06
```

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